Note Detection based on Song Attributes

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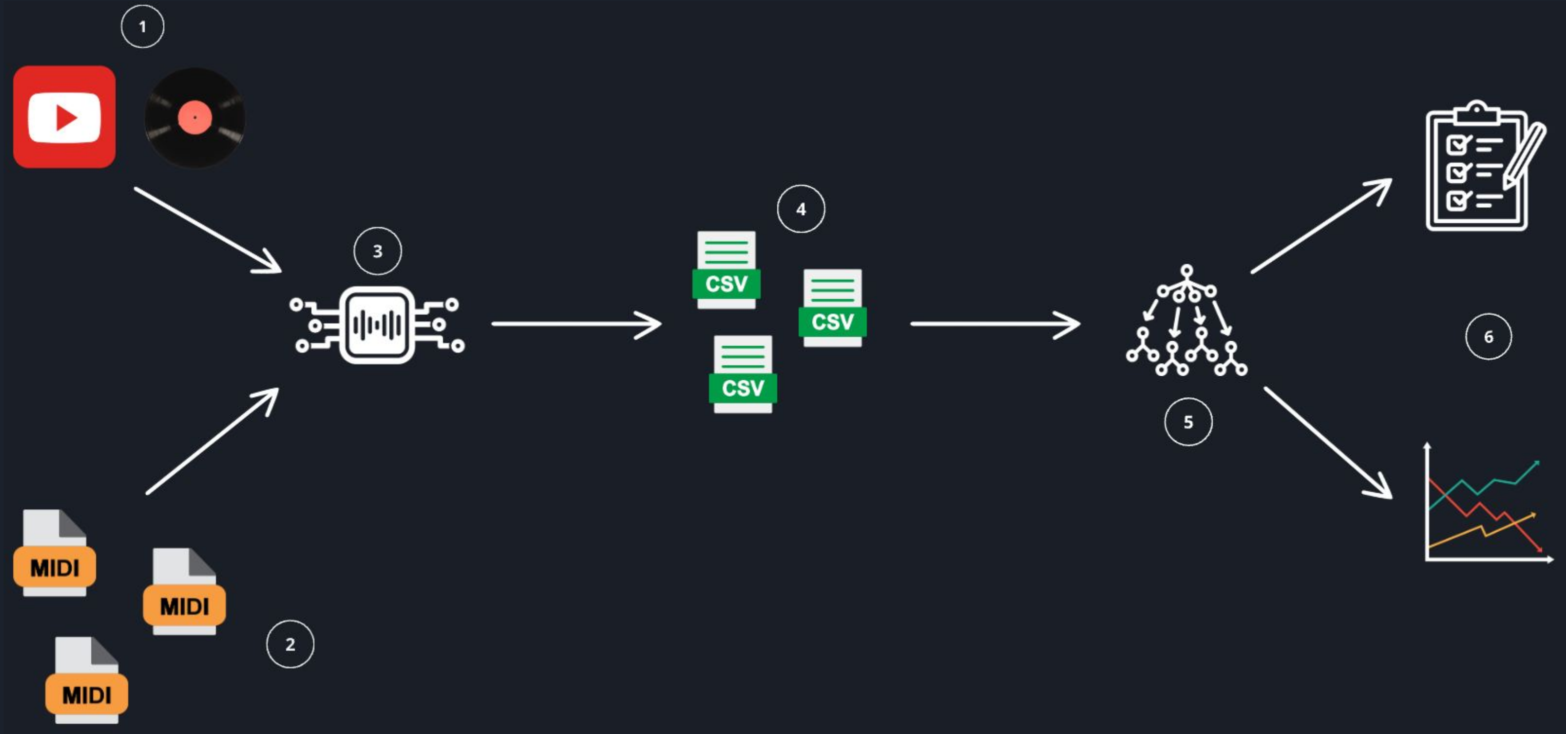
Agenda 🔖

- Scope
- Methodology
- Data Acquisition
- Data Prep
- Model Tuning
    - SVM
    - Random Forests
    - XG Boost
    - LSTM
- Comparison
- Key Findings & Future Work

Scope 🧐

Methodology 🤓



Data Acquisition ⛏️

1. YT Audio Download & Conversion
   a. yt_dlp
   b. mainly Karaoke Versions
   c. .wav format
   d. extract features using librosa
      i. time frames
      ii. pitch
      iii. power
      iv. spectral centroid
2. MIDI extract
   a. .mid files through web (https://onlinesequencer.net/)
   b. mido library
      i. Time
      ii. MIDI_NOTE

Data Prep 🧹

- Alignment MIDI-Audio
  - Match MIDI note with time frame
  - Calculate Prv Note (on Load)
  - Merge into One dataframe
- CSV Creation

aligned_features_Queen - I Want it All.csv

```
Time,MIDI_Note,Note_Label,Pitch,Power,Spectral_Centroid
1.6304325,64,E4,1148.35791015625,0.034626174718141556,2679.3498771550408
1.6304325,60,C4,1148.35791015625,0.034626174718141556,2679.3498771550408
1.6304325,57,A3,1148.35791015625,0.034626174718141556,2679.3498771550408
1.956519,64,E4,1159.25048828125,0.10618942230939865,2245.739777755514
1.956519,60,C4,1159.25048828125,0.10618942230939865,2245.739777755514
1.956519,57,A3,1159.25048828125,0.10618942230939865,2245.739777755514
2.2826055,64,E4,1160.6787109375,0.09879202395677567,2211.1102872730867
2.2826055,60,C4,1160.6787109375,0.09879202395677567,2211.1102872730867
2.2826055,57,A3,1160.6787109375,0.09879202395677567,2211.1102872730867
```

ML Models & Tuning 🤖

- Models Tested
  - **XGBoost:** Ability to handle class imbalance well
  - **Random Forest:** Core ML baseline for comparison (also one of my favourite 😃).
  - **SVM:** Effective for classification tasks
  - **LSTM:** Sequence-learning capability.
- Hyper-Parameter Tuning
  - GridSearchCV (too long 🥱)
  - RandomizedSearchCV (better times 😃)
  - Optuna (Speed & Efficiency 🚀)

First Results 😫

🔹 XGBoost Evaluation Metrics:
✅ Accuracy: 0.0685
✅ Precision: 0.0743
✅ Recall: 0.0685
✅ F1 Score: 0.0656
🎵 Mean Absolute Error (Note Distance): 10.9973
🎯 Top-3 Accuracy: 0.1532
⏳ Training Time: 7.9776 sec
⚡ Prediction Time: 0.0610 sec
💾 Model Size: 0.00 MB

🔹 SVM Evaluation Metrics:
✅ Accuracy: 0.0353
✅ Precision: 0.0040
✅ Recall: 0.0353
✅ F1 Score: 0.0072
🎵 Mean Absolute Error (Note Distance): 11.2518
⏳ Training Time: 55.6013 sec
⚡ Prediction Time: 10.5257 sec
💾 Model Size: 0.00 MB

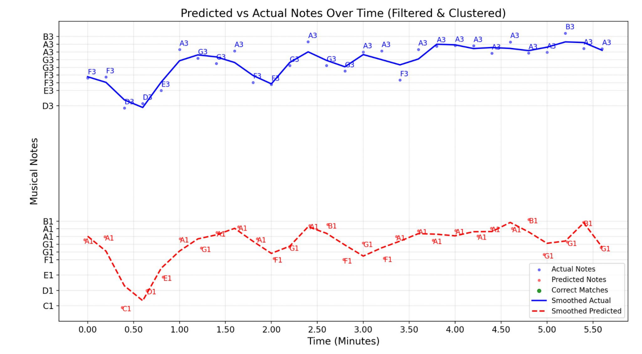
🔹 Random Forest Evaluation Metrics:
✅ Accuracy: 0.0767
✅ Precision: 0.1087
✅ Recall: 0.0767
✅ F1 Score: 0.0740
🎵 Mean Absolute Error (Note Distance): 11.0335
🎯 Top-3 Accuracy: 0.1691
⏳ Training Time: 0.6721 sec
⚡ Prediction Time: 0.0363 sec
💾 Model Size: 0.00 MB

🔹 LSTM Evaluation Metrics:
✅ Accuracy: 0.0312
✅ Precision: 0.0017
✅ Recall: 0.0312
✅ F1 Score: 0.0030
🎵 Mean Absolute Error (Note Distance): 10.6018
⏳ Training Time: 0.0000 sec
⚡ Prediction Time: 0.1532 sec
💾 Model Size: 0.00 MB

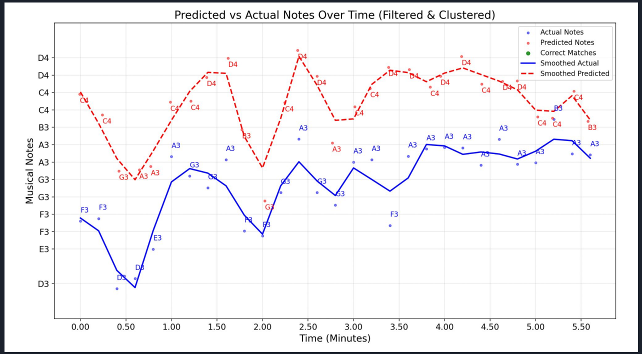
Side to Side Comparison

Model	✅ Accuracy	🎯 Precision	🔄 Recall	📊 F1 Score	🎵 MAE (Note Distance)	🏆 Top-3 Accuracy	⏳ Training Time (sec)	⚡ Prediction Time (sec)
🔹 **SVM**	0.0353	4	0.0353	0.0072	11.2518	N/A	55.6013	10.5257
🚀 **XGBoost**	0.0685	0.0743	0.0685	0.0656	10.9973	0.1532	7.9776	61
🧠 **LSTM**	0.0312	0.0017	0.0312	3	⏳ 10.6018	N/A	⏳ 0.0	0.1532
🌲 **Random Forest**	✅ 0.0767	✅ 0.1087	✅ 0.0767	✅ 0.074	11.0335	✅ 0.1691	0.6721	⏳ 0.0363

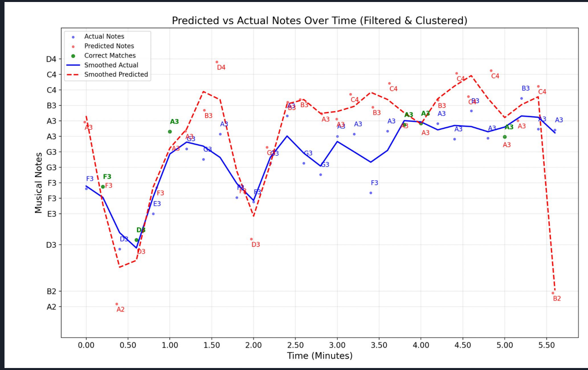
A Different Approach 🥹



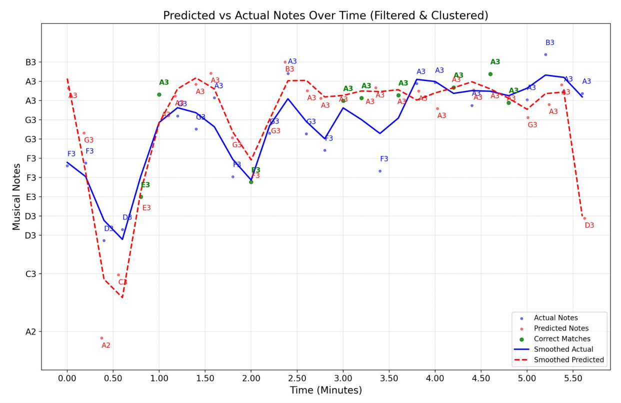
XG Boost



Random Forest



SVM



LSTM

Key Findings 💡

- No “One to Rule them All” Solution
- Data Acquisition & Cleaning.. a Necessary Headache
- Even with Hyperparameter Tuning.. each model has each own Strengths and Weaknesses
- Evaluation Metrics can be Deceiving Sometimes

Ideas for Further Work

- Training Set Enhancement
- More Advanced Tuning
- Individual Tests
  - Per Genre
  - Per Type (i.e instrumental, vocals-only)
- Web-App for Applying the Model in user-given Songs
- Real-Time Note-Identification

Any Questions? 🤔

Thank you for your Time